Chapter 4. Double Integrals

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Overview

Results from Chapters 1–3 are cited as *Fact B.k*; they are stated in the separate *Background* notes. When dealing with functions of several variables one constantly needs to interchange the order of integration. To do this with the Lebesgue integral we must first have a measure on the plane. This chapter defines the **product** of two general measures, uses it to define double integrals, and determines when a double integral equals the iterated integrals (Tonelli, Fubini). Double integrals are essential for defining the **convolution**, which appears constantly in Fourier analysis.

4.1 Product Measures

4.1.1 Setting up the problem

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be measure spaces. To put a measure on $X \times Y$ we first need a σ-algebra, which should certainly contain the **measurable rectangles**

$$\mathcal{R} = \{S \times T \subset X \times Y : \ S \in \mathcal{S}, \ T \in \mathcal{T}\}.$$

Once a σ-algebra is fixed, the way to define the measure is clear. For $E \subset X \times Y$, $x \in X$, $y \in Y$ define the **sections**

$$E_x = \{y \in Y : (x, y) \in E\}, \qquad E^y = \{x \in X : (x, y) \in E\}.$$

If $E_x \in \mathcal{T}$ and $E^y \in \mathcal{S}$ for all x, y, we can define

$$\varphi : x \mapsto \nu(E_x) = \int_Y \chi_E(x, y)\, d\nu(y), \qquad \psi : y \mapsto \mu(E^y) = \int_X \chi_E(x, y)\, d\mu(x). \tag{1}$$

If $\varphi : X \to [0, \infty]$ and $\psi : Y \to [0, \infty]$ are measurable and

$$\int_X \varphi \, d\mu = \int_Y \psi \, d\nu, \tag{2}$$

it is natural to define this common value to be the measure of E. The conclusion will be: if $\mathcal{S} \times \mathcal{T}$ denotes the smallest σ-algebra containing $\mathcal{R}$, every $E \in \mathcal{S} \times \mathcal{T}$ has all these properties.

4.1.2 The smallest σ-algebra

Let $\mathcal{U}$ be any collection of subsets of a set Z and let Ω be the family of all σ-algebras containing $\mathcal{U}$; $\Omega \neq \emptyset$ since the power set belongs to it. Put

$$\mathcal{W} = \bigcap \{\mathcal{M} : \mathcal{M} \in \Omega\}.$$

Clearly $\emptyset \in \mathcal{W}$. If $A \in \mathcal{W}$ then $A \in \mathcal{M}$ for every $\mathcal{M} \in \Omega$, hence $A^c \in \mathcal{M}$ for every $\mathcal{M}$, so $A^c \in \mathcal{W}$. Likewise $A_n \in \mathcal{W}$ $(n = 1, 2, \dots)$ implies $\bigcup_n A_n \in \mathcal{W}$. Thus $\mathcal{W}$ is a σ-algebra, and if $\mathcal{M}$ is any σ-algebra containing $\mathcal{U}$ then $\mathcal{W} \subset \mathcal{M}$: $\mathcal{W}$ is the **smallest σ-algebra containing $\mathcal{U}$**.

4.1.3 Sections of sets in $\mathcal{S} \times \mathcal{T}$

To show $E_x \in \mathcal{T}$ for every $E \in \mathcal{S} \times \mathcal{T}$ and $x \in X$, let

$$\mathcal{A} = \{E \in \mathcal{S} \times \mathcal{T} : \ E_x \in \mathcal{T} \ (x \in X)\}.$$

Clearly $\mathcal{R} \subset \mathcal{A}$, so it suffices to show $\mathcal{A}$ is a σ-algebra (then $\mathcal{A} \supset \mathcal{S} \times \mathcal{T}$). $\emptyset \in \mathcal{A}$. If $E \in \mathcal{A}$ then $(E^c)_x = (E_x)^c$, so $E^c \in \mathcal{A}$. If $E_n \in \mathcal{A}$ then $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$, so $\bigcup_n E_n \in \mathcal{A}$. Treating E^y the same way gives

$$E \in \mathcal{S} \times \mathcal{T} \implies E_x \in \mathcal{T}, \ E^y \in \mathcal{S} \qquad (x \in X, \ y \in Y). \tag{3}$$

Exercise 4.1.1

In the argument above, show that $(E^c)_x = (E_x)^c$ and $(\bigcup_n E_n)_x = \bigcup_n (E_n)_x$.

Solution. $y \in (E^c)_x \iff (x,y) \in E^c \iff (x,y) \notin E \iff y \notin E_x \iff y \in (E_x)^c$. $y \in (\bigcup_n E_n)_x \iff (x,y) \in E_n$ for some $n \iff y \in (E_n)_x$ for some $n \iff y \in \bigcup_n (E_n)_x$. □

4.1.4 The monotone class argument

From now on assume X and Y can be partitioned into sets of finite measure:

$$X = \bigsqcup_{n=1}^{\infty} X_n, \qquad Y = \bigsqcup_{m=1}^{\infty} Y_m, \qquad \mu(X_n),\ \nu(Y_m) < \infty.$$

To show that every $E \in \mathcal{S} \times \mathcal{T}$ satisfies (2), consider

$$\mathcal{B} = \{E \in \mathcal{S} \times \mathcal{T} :\ \text{the functions } \varphi, \psi \text{ of (1) are measurable and (2) holds}\},$$
$$\mathcal{C} = \{E \in \mathcal{S} \times \mathcal{T} :\ E \cap (X_n \times Y_m) \in \mathcal{B}\ \ (m, n = 1, 2, \dots)\}.$$

Lemma 4.1.1

For the collections $\mathcal{B}$ and $\mathcal{C}$:

(a) $\mathcal{B}$ is closed under unions of increasing sequences.

(b) $\mathcal{B}$ is closed under countable disjoint unions.

(c) $\mathcal{C}$ is closed under unions of increasing sequences and intersections of decreasing sequences.

(d) If $E_1, \dots, E_N \in \mathcal{R}$ are pairwise disjoint, then $\bigcup_{n=1}^{N} E_n \in \mathcal{C}$.

Proof. (a) Let $\langle E_i \rangle$ be an increasing sequence in $\mathcal{B}$ with union E, and let φ_i, ψ_i and φ, ψ be the corresponding functions in (1). Since $\langle (E_i)_x \rangle$ increases to E_x, Fact B.24(b) (continuity from below) gives $\varphi_i \nearrow \varphi$. Each φ_i is measurable, so φ is measurable; the same holds for ψ_i, ψ. By the monotone convergence theorem,

$$\int_X \varphi \, d\mu = \lim_{i\to\infty} \int_X \varphi_i \, d\mu = \lim_{i\to\infty} \int_Y \psi_i \, d\nu = \int_Y \psi \, d\nu.$$

(b) If $E, F \in \mathcal{B}$ are disjoint, the functions in (1) for $E \sqcup F$ are sums of those for E and F, so $E \sqcup F \in \mathcal{B}$; by induction $\mathcal{B}$ is closed under finite disjoint unions. A countable disjoint union is the increasing union of the finite partial unions, so (a) applies.

(c) If $\langle E_i \rangle$ is an increasing sequence in $\mathcal{C}$, then for each m, n the sequence

$$\langle E_i \cap (X_n \times Y_m) : i = 1, 2, \dots \rangle \tag{4}$$

is an increasing sequence in $\mathcal{B}$, so by (a) $\bigcup_i E_i \in \mathcal{C}$. If $\langle E_i \rangle$ is decreasing in $\mathcal{C}$, consider the intersection of the sequence (4). Proceed as in (a), but use Fact B.24(c) (continuity from above, valid because $\nu(Y_m), \mu(X_n) < \infty$) and the Lebesgue (dominated) convergence theorem (the functions are bounded by $\nu(Y_m)\chi_{X_n}$, which is integrable). Hence the intersection lies in $\mathcal{B}$, proving (c).

(d) Clearly $\mathcal{R} \subset \mathcal{C}$: for $E = S \times T$, $\varphi = \nu(T \cap Y_m)\chi_{S \cap X_n}$ and $\psi = \mu(S \cap X_n)\chi_{T \cap Y_m}$ are measurable with equal integrals $\mu(S \cap X_n)\nu(T \cap Y_m)$. Then (b) gives (d). ■

Now let $\mathcal{D}$ be the smallest collection containing $\mathcal{R}$ and satisfying (c) and (d) of Lemma 4.1.1. Then

$$\mathcal{D} \subset \mathcal{C} \subset \mathcal{S} \times \mathcal{T}, \tag{5}$$

and we will show that $\mathcal{D}$ is a σ-algebra. Let $\mathcal{E}$ be the collection of sets expressible as finite disjoint unions of measurable rectangles. Then $\mathcal{D}$ is the smallest collection containing $\mathcal{E}$ and closed under unions of increasing sequences and intersections of decreasing sequences (a *monotone class*).

Exercise 4.1.2

Show that for any collection of sets there is a smallest collection containing it that is closed under unions of increasing sequences and intersections of decreasing sequences.

Solution. Let $\mathcal{U}$ be a collection of subsets of Z and Ω the family of all monotone classes containing $\mathcal{U}$ ($\Omega \neq \emptyset$: the power set of Z is one). The intersection $\mathcal{W} = \bigcap \Omega$ contains $\mathcal{U}$. If $\langle A_i \rangle$ is an increasing (decreasing) sequence in $\mathcal{W}$, then it is such a sequence in every $\mathcal{M} \in \Omega$, so $\bigcup A_i$ ($\bigcap A_i$) lies in every $\mathcal{M}$, hence in $\mathcal{W}$. So $\mathcal{W}$ is a monotone class, and it is contained in every monotone class containing $\mathcal{U}$. □

A simple set computation shows that $\mathcal{E}$ is closed under differences and unions:

$$E, F \in \mathcal{E} \implies E \setminus F \in \mathcal{E},\ F \setminus E \in \mathcal{E},\ E \cup F \in \mathcal{E}. \tag{6}$$

Exercise 4.1.3

Prove (6).

Solution. (i) The intersection of two rectangles is a rectangle: $(S \times T) \cap (S' \times T') = (S \cap S') \times (T \cap T')$. Hence if $E = \bigsqcup_i R_i$, $F = \bigsqcup_j R'_j$ are in $\mathcal{E}$, then $E \cap F = \bigsqcup_{i,j} (R_i \cap R'_j) \in \mathcal{E}$; so $\mathcal{E}$ is closed under finite intersections. (ii) The complement of a rectangle is a disjoint union of two rectangles: $(S \times T)^c = (S^c \times Y) \sqcup (S \times T^c)$. So $(\bigsqcup_i R_i)^c = \bigcap_i R_i^c$ is a finite intersection of members of $\mathcal{E}$, hence in $\mathcal{E}$ by (i). (iii) Therefore $E \setminus F = E \cap F^c \in \mathcal{E}$, likewise $F \setminus E \in \mathcal{E}$, and $E \cup F = (E \setminus F) \sqcup F \in \mathcal{E}$, the union being disjoint and each piece a finite disjoint union of rectangles. □

For an arbitrary $E \subset X \times Y$ define

$$\Omega(E) = \{F \subset X \times Y :\ E \setminus F \in \mathcal{D},\ F \setminus E \in \mathcal{D},\ E \cup F \in \mathcal{D}\}.$$

Then $\Omega(E)$ also satisfies Lemma 4.1.1(c).

Exercise 4.1.4

Show that $\Omega(E)$ satisfies (c) of Lemma 4.1.1.

Solution. Let $F_i \in \Omega(E)$ increase to F. Then $\langle E \setminus F_i \rangle$ decreases to $E \setminus F$, $\langle F_i \setminus E \rangle$ increases to $F \setminus E$, and $\langle E \cup F_i \rangle$ increases to $E \cup F$; all terms lie in $\mathcal{D}$, which is a monotone class, so the three limits lie in $\mathcal{D}$ and $F \in \Omega(E)$. If F_i decreases to F, then $E \setminus F_i \nearrow E \setminus F$, $F_i \setminus E \searrow F \setminus E$, $E \cup F_i \searrow E \cup F$, and the same argument applies. □

If $E, F \in \mathcal{E}$, then by (6) $F \in \Omega(E)$, so $\mathcal{E} \subset \Omega(E)$ and therefore $\mathcal{D} \subset \Omega(E)$, i.e.

$$E \in \mathcal{E} \implies \mathcal{D} \subset \Omega(E).$$

Hence if $E \in \mathcal{D}$ and $F \in \mathcal{E}$, then $E \in \Omega(F)$, which is equivalent to $F \in \Omega(E)$ (the definition is symmetric). So $\mathcal{E} \subset \Omega(E)$ and thus $\mathcal{D} \subset \Omega(E)$. In other words

$$E \in \mathcal{D} \implies \mathcal{D} \subset \Omega(E),$$

which says exactly that $\mathcal{D}$ is closed under differences and finite unions. Since $X \times Y \in \mathcal{D}$, $\mathcal{D}$ is closed under complements. If $E_n \in \mathcal{D}$, put $F_n = E_1 \cup \cdots \cup E_n$; then $\langle F_n \rangle$ is increasing in $\mathcal{D}$, so $\bigcup_n E_n = \bigcup_n F_n \in \mathcal{D}$. Thus $\mathcal{D}$ is a σ-algebra. Since $\mathcal{S} \times \mathcal{T}$ is the smallest σ-algebra containing $\mathcal{E}$, (5) gives

$$\mathcal{D} = \mathcal{C} = \mathcal{S} \times \mathcal{T}. \tag{7}$$

Consequently, if $E \in \mathcal{S}\times\mathcal{T} = \mathcal{C}$, then $E\cap(X_n\times Y_m) \in \mathcal{B}$ for all m,n; these sets are pairwise disjoint with union E, so $E\in\mathcal{B}$ by Lemma 4.1.1(b). We have shown $\mathcal{S}\times\mathcal{T}\subset\mathcal{B}$.

Definition (σ-finite)

A measure space $(X,\mathcal{S},\mu)$ is **σ-finite** if X is a countable union of sets of finite measure.

Theorem 4.1.2

Let $(X,\mathcal{S},\mu)$, $(Y,\mathcal{T},\nu)$ be measure spaces and let $\mathcal{S}\times\mathcal{T}$ be the smallest σ-algebra containing the measurable rectangles. If $E\in\mathcal{S}\times\mathcal{T}$, then $E_x\in\mathcal{T}$ and $E^y\in\mathcal{S}$ for all $x\in X$, $y\in Y$. If $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ are σ-finite, then the functions φ,ψ of (1) are measurable and (2) holds.

(The proof is the discussion above.) Writing the common value of (2) as $(\mu\times\nu)(E)$,

$$\begin{aligned}(\mu\times\nu)(E) &= \int_X \Big[\int_Y \chi_E(x,y)\,d\nu(y)\Big]d\mu(x) = \int_X \nu(E_x)\,d\mu(x)\\ &= \int_Y \Big[\int_X \chi_E(x,y)\,d\mu(x)\Big]d\nu(y) = \int_Y \mu(E^y)\,d\nu(y).\end{aligned} \tag{8}$$

$\mu\times\nu$ is a measure: if $E_n\in\mathcal{S}\times\mathcal{T}$ are disjoint with $E=\bigsqcup_n E_n$, then $E_x=\bigsqcup_n (E_n)_x$, and by Fact B.25 (term-by-term integration of nonnegative series),

$$\sum_{n=1}^{\infty}(\mu\times\nu)(E_n) = \sum_{n=1}^{\infty}\int_X \nu((E_n)_x)\,d\mu(x) = \int_X \sum_{n=1}^{\infty}\nu((E_n)_x)\,d\mu(x) = \int_X \nu(E_x)\,d\mu(x) = (\mu\times\nu)(E).$$

The measure space $(X\times Y,\mathcal{S}\times\mathcal{T},\mu\times\nu)$ is the **product measure** of the σ-finite measure spaces $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$.

Exercise 4.1.5

Show that $(\mu\times\nu)(S\times T)=\mu(S)\nu(T)$ for every $S\in\mathcal{S}$, $T\in\mathcal{T}$.

Solution. $(S\times T)_x = T$ if $x\in S$ and $=\emptyset$ otherwise, so $\nu((S\times T)_x)=\nu(T)\chi_S(x)$. By (8), $(\mu\times\nu)(S\times T)=\int_X \nu(T)\chi_S\,d\mu=\nu(T)\mu(S)$ (with $0\cdot\infty=0$). □

Exercise 4.1.6

If $(X,\mathcal{S},\mu)$ and $(Y,\mathcal{T},\nu)$ are σ-finite, show that $(X\times Y,\mathcal{S}\times\mathcal{T},\mu\times\nu)$ is σ-finite.

Solution. With $X=\bigsqcup_n X_n$, $Y=\bigsqcup_m Y_m$ as above, $X\times Y=\bigsqcup_{m,n}(X_n\times Y_m)$ is a countable union, and by Exercise 4.1.5 $(\mu\times\nu)(X_n\times Y_m)=\mu(X_n)\nu(Y_m)<\infty$. □

4.1.5 The product of Lebesgue measures; Borel sets

Consider the product of Lebesgue measure $(\mathbb{R},\mathcal{M},m)$ with itself. Given an open set $U\subset\mathbb{R}^2$, the collection

$$\{(a,b)\times(c,d)\subset U:\ a,b,c,d \text{ rational}\}$$

is countable and its union is U, so $U\in\mathcal{M}\times\mathcal{M}$: every open set is in $\mathcal{M}\times\mathcal{M}$. Let $\mathcal{B}_n$ be the smallest σ-algebra containing the open sets of $\mathbb{R}^n$. Then

$$\mathcal{B}_2\subset\mathcal{M}\times\mathcal{M}.$$

Members of $\mathcal{B}_n$ are called **Borel sets** of $\mathbb{R}^n$; closed sets, G_δ-sets and F_σ-sets are typical examples.

However, $\mathcal{M}\times\mathcal{M}$ has a defect. Take $E\in\mathcal{M}$ with $m(E)=0$, $E\neq\emptyset$, and $F\notin\mathcal{M}$, $F\subset\mathbb{R}$. By (3), $E\times F\notin\mathcal{M}\times\mathcal{M}$ (its sections $(E\times F)^y=E$ for $y\in F$, but $(E\times F)_x=F\notin\mathcal{M}$ for $x\in E$). Yet $E\times F\subset E\times\mathbb{R}$ and $(m\times m)(E\times\mathbb{R})=0$. So a subset of a null set can fail to be measurable.

Definition (complete measure)

A measure space $(X,\mathcal{S},\mu)$ is **complete** if

$$E\subset F,\ F\in\mathcal{S},\ \mu(F)=0\implies E\in\mathcal{S}.$$

Thus $(\mathbb{R}^2,\mathcal{M}\times\mathcal{M},m\times m)$ is not complete, while $(\mathbb{R},\mathcal{M},m)$ is complete by Fact B.2.

Proposition 4.1.3 (completion)

For a measure space $(X,\mathcal{S},\mu)$ let

$$\mathcal{S}^*=\{E\subset X:\ \text{there exist } A,B\in\mathcal{S}\text{ with } A\subset E\subset B,\ \mu(B\setminus A)=0\},$$

and for $E\in\mathcal{S}^*$ put $\mu^*(E)=\mu(A)$. Then $(X,\mathcal{S}^*,\mu^*)$ is a complete measure space.

Proof. *Well-definedness.* Suppose $A_i\subset E\subset B_i$, $\mu(B_i\setminus A_i)=0$ for $i=1,2$. Then

$$A_2\setminus A_1\subset E\setminus A_1\subset B_1\setminus A_1,$$

so $\mu(A_2\setminus A_1)=0$; likewise $\mu(A_1\setminus A_2)=0$, hence $\mu(A_1)=\mu(A_2)$.

$\mathcal{S}^*$ is a σ-algebra. Clearly $\emptyset\in\mathcal{S}^*$. If $E\in\mathcal{S}^*$ with $A\subset E\subset B$, $\mu(B\setminus A)=0$, then $B^c\subset E^c\subset A^c$ and $A^c\setminus B^c=B\setminus A$, so $E^c\in\mathcal{S}^*$. Finally let $A_n\subset E_n\subset B_n$ with $\mu(B_n\setminus A_n)=0$, and put $A=\bigcup_n A_n$, $E=\bigcup_n E_n$, $B=\bigcup_n B_n$. Then $A\subset E\subset B$ and

$$B\setminus A=\bigcup_n(B_n\setminus A)\subset\bigcup_n(B_n\setminus A_n),$$

so $\mu(B\setminus A)=0$ and $E\in\mathcal{S}^*$.

μ^* is a measure. If the E_n are disjoint, so are the A_n, and

$$\mu^*(E)=\mu(A)=\sum_{n=1}^{\infty}\mu(A_n)=\sum_{n=1}^{\infty}\mu^*(E_n).$$

Completeness. Let $E\subset F$, $F\in\mathcal{S}^*$, $\mu^*(F)=0$. Choose $A,B\in\mathcal{S}$ with $A\subset F\subset B$ and $\mu(A)=\mu(B)=0$. Then $\emptyset\subset E\subset B$ and $\mu(B)=0$, so $E\in\mathcal{S}^*$. ∎

Exercise 4.1.7

If $(X,\mathcal{S},\mu)$ is complete, show that $\mathcal{S}^*=\mathcal{S}$.

Solution. $\mathcal{S}\subset\mathcal{S}^*$ always (take $A=B=E$). Conversely, let $E\in\mathcal{S}^*$ with $A\subset E\subset B$, $\mu(B\setminus A)=0$. Then $E\setminus A\subset B\setminus A$, a null set in $\mathcal{S}$, so $E\setminus A\in\mathcal{S}$ by completeness, and $E=A\cup(E\setminus A)\in\mathcal{S}$. □

4.1.6 Lebesgue measure on $\mathbb{R}^2$

Compare the two measures $(\mathbb{R}^2,\mathcal{B}_2,m\times m)$ and $(\mathbb{R}^2,\mathcal{M}\times\mathcal{M},m\times m)$, i.e. $\mathcal{B}_2^*$ and $(\mathcal{M}\times\mathcal{M})^*$. Since $\mathcal{B}_2\subset\mathcal{M}\times\mathcal{M}$, clearly $\mathcal{B}_2^*\subset(\mathcal{M}\times\mathcal{M})^*$.

If $S\in\mathcal{M}$, Fact B.3 gives an F_σ-set A and a G_δ-set B with $A\subset S\subset B$, $m(B\setminus A)=0$. Then $A\times\mathbb{R}$ and $B\times\mathbb{R}$ are F_σ- and G_δ-sets of $\mathbb{R}^2$, so they lie in $\mathcal{B}_2$, and

$$(m\times m)[(B\times\mathbb{R})\setminus(A\times\mathbb{R})]=(m\times m)[(B\setminus A)\times\mathbb{R}]=0,$$

so $S \times \mathbb{R} \in \mathcal{B}_2^*$. Similarly $T \in \mathcal{M}$ implies $\mathbb{R} \times T \in \mathcal{B}_2^*$, so every measurable rectangle $S \times T = (S \times \mathbb{R}) \cap (\mathbb{R} \times T)$ belongs to $\mathcal{B}_2^*$. Hence $\mathcal{M} \times \mathcal{M} \subset \mathcal{B}_2^*$, and since $(\mathcal{M} \times \mathcal{M})^* \subset \mathcal{B}_2^{**} = \mathcal{B}_2^*$,

$$(\mathcal{M} \times \mathcal{M})^* = \mathcal{B}_2^*.$$

Write $\mathcal{M}_2$ for this σ-algebra and m_2 for the measure $(m \times m)^*$ on it. The complete measure space $(\mathbb{R}^2, \mathcal{M}_2, m_2)$ is the **Lebesgue measure on** $\mathbb{R}^2$. (If one only wants Lebesgue measure on $\mathbb{R}^n$ from scratch, the same method used for $\mathbb{R}^1$ applies.)

Exercise 4.1.8

Use induction to define Lebesgue measure on $\mathbb{R}^n$.

Solution. Suppose the complete measure space $(\mathbb{R}^{n-1}, \mathcal{M}_{n-1}, m_{n-1})$ has been defined ($m_1 = m$); it is σ-finite since $\mathbb{R}^{n-1} = \bigcup_k [-k, k]^{n-1}$. Identify $\mathbb{R}^n = \mathbb{R}^{n-1} \times \mathbb{R}$, form the product measure $(\mathbb{R}^n, \mathcal{M}_{n-1} \times \mathcal{M}, m_{n-1} \times m)$ (Theorem 4.1.2), and define

$$\mathcal{M}_n = (\mathcal{M}_{n-1} \times \mathcal{M})^*, \qquad m_n = (m_{n-1} \times m)^*$$

(Proposition 4.1.3). Exactly as above one checks $\mathcal{B}_n \subset \mathcal{M}_{n-1} \times \mathcal{M}$ (open sets are countable unions of rational boxes) and $\mathcal{M}_n = \mathcal{B}_n^*$, using inductively that each $S \in \mathcal{M}_{n-1}$ is squeezed between an F_σ- and a G_δ-set of $\mathbb{R}^{n-1}$ differing by a null set. By Exercise 4.1.5 and induction, m_n of a box $\prod_i [a_i, b_i]$ is $\prod_i (b_i - a_i)$. □

4.2 Fubini's Theorem

Equation (8) says that the order of integration can be interchanged for characteristic functions of sets in $\mathcal{S} \times \mathcal{T}$. Since the integral is built from integrals of characteristic functions, we expect the same for nonnegative measurable functions and for integrable functions. For a function f on $X \times Y$ define

$$f_x : y \mapsto f(x, y), \qquad f^y : x \mapsto f(x, y).$$

Proposition 4.2.1

If $f : X \times Y \to \mathbb{R}^*$ is $\mathcal{S} \times \mathcal{T}$-measurable, then for every $x \in X$ and $y \in Y$, f_x is $\mathcal{T}$-measurable and f^y is $\mathcal{S}$-measurable.

Proof. For $\alpha \in \mathbb{R}$ put $E = \{(x, y) \in X \times Y : f(x, y) > \alpha\}$. Then $E_x = \{y \in Y : f_x(y) > \alpha\}$. Since $E \in \mathcal{S} \times \mathcal{T}$, (3) gives $E_x \in \mathcal{T}$, so f_x is measurable. Similarly f^y is measurable. ■

From now on, "measurable" on $X \times Y$ means measurable with respect to $\mathcal{S} \times \mathcal{T}$.

Exercise 4.2.1

If $f : \mathbb{R}^2 \to \mathbb{R}$ is measurable with respect to Lebesgue measure on $\mathbb{R}^2$, is $f_x : \mathbb{R} \to \mathbb{R}$ always Lebesgue measurable for every $x \in \mathbb{R}$?

Solution. No. Let $N \subset \mathbb{R}$ be a non-measurable set and $f = \chi_{\{0\} \times N}$. Since $\{0\} \times N \subset \{0\} \times \mathbb{R}$ and $m_2(\{0\} \times \mathbb{R}) = m(\{0\}) m(\mathbb{R}) = 0$, completeness gives $\{0\} \times N \in \mathcal{M}_2$, so f is $\mathcal{M}_2$-measurable. But $f_0 = \chi_N$ is not Lebesgue measurable. (What is true is that f_x is measurable for *almost every* x; see Theorem 4.2.6.) □

Theorem 4.2.2 (Tonelli)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite. For every measurable $f : X \times Y \to [0, \infty]$:

(a) the functions $x \mapsto \int_Y f_x \, d\nu$ and $y \mapsto \int_X f^y \, d\mu$ are measurable;

(b)
$$\int_X \Big[\int_Y f_x \, d\nu \Big] d\mu = \int_{X \times Y} f \, d(\mu \times \nu) = \int_Y \Big[\int_X f^y \, d\mu \Big] d\nu.$$

Proof. By Fact B.5 there is an increasing sequence of simple functions $\langle s_n \rangle$ with $s_n \nearrow f$ pointwise. Put
$$\varphi(x) = \int_Y f_x \, d\nu, \qquad \varphi_n(x) = \int_Y (s_n)_x \, d\nu.$$
By Theorem 4.1.2, characteristic functions of measurable sets satisfy (a) and (b), and by linearity so do simple functions. Since $s_n \nearrow f$, we have $(s_n)_x \nearrow f_x$, and the monotone convergence theorem for ν gives $\varphi_n \nearrow \varphi$. Hence φ is measurable. Applying the monotone convergence theorem for $\mu \times \nu$ and for μ,
$$\int_{X \times Y} f \, d(\mu \times \nu) = \lim_{n \to \infty} \int_{X \times Y} s_n \, d(\mu \times \nu) = \lim_{n \to \infty} \int_X \varphi_n \, d\mu = \int_X \varphi \, d\mu.$$
The other half is proved in exactly the same way. ■

Exercise 4.2.2

Derive Fact B.25 from Tonelli's theorem.

Solution. Formula Fact B.25 is term-by-term integration: for measurable $f_n : X \to [0, \infty]$, $\int_X \sum_n f_n \, d\mu = \sum_n \int_X f_n \, d\mu$. Let $Y = \mathbb{N}$ with the counting measure c (on the power set), which is σ-finite, and define $f(x, n) = f_n(x)$. Every subset of $\mathbb{N}$ is measurable, and $\{f > \alpha\} = \bigcup_n \{f_n > \alpha\} \times \{n\}$, so f is $\mathcal{S} \times \mathcal{P}(\mathbb{N})$-measurable. Since $\int_{\mathbb{N}} g \, dc = \sum_n g(n)$, Tonelli gives
$$\int_X \sum_n f_n(x) \, d\mu(x) = \int_{X \times \mathbb{N}} f \, d(\mu \times c) = \sum_n \int_X f_n \, d\mu.$$
□

The same holds for integrable functions; we first find a sufficient condition for f to be integrable.

Theorem 4.2.3 (Fubini)

Let $(X, \mathcal{S}, \mu)$ and $(Y, \mathcal{T}, \nu)$ be σ-finite and let $f : X \times Y \to \mathbb{C}$ be measurable with
$$\int_X \Big[\int_Y |f(x, y)| \, d\nu(y) \Big] d\mu(x) < \infty.$$
Then:

(a) $f \in L^1(\mu \times \nu)$; moreover $f_x \in L^1(\nu)$ for almost every $x \in X$ and $f^y \in L^1(\mu)$ for almost every $y \in Y$;

(b) the functions $x \mapsto \int_Y f_x \, d\nu$ and $y \mapsto \int_X f^y \, d\mu$, defined almost everywhere, are integrable;

(c)
$$\int_X \Big[\int_Y f_x \, d\nu \Big] d\mu = \int_{X \times Y} f \, d(\mu \times \nu) = \int_Y \Big[\int_X f^y \, d\mu \Big] d\nu.$$

Proof. By Tonelli, $\int_{X\times Y}|f| = \int_X[\int_Y |f(x,y)|\,d\nu(y)]d\mu(x) < \infty$, so $f \in L^1(\mu\times\nu)$. Writing $f = u+iv$, it suffices to treat $f : X \times Y \to \mathbb{R}$. Put

$$\varphi(x) = \int_Y f_x\,d\nu, \qquad \varphi_1(x) = \int_Y (f_+)_x\,d\nu, \qquad \varphi_2(x) = \int_Y (f_-)_x\,d\nu.$$

By Tonelli,

$$\int_X \varphi_1\,d\mu = \int_{X\times Y} f_+\,d(\mu\times\nu) < \infty,$$

so $\varphi_1 \in L^1(\mu)$, and likewise $\varphi_2 \in L^1(\mu)$. Therefore the set

$$E = \{x \in X : \varphi_1(x) < \infty,\ \varphi_2(x) < \infty\}$$

satisfies $\mu(E^c) = 0$. For each $x \in E$, $f_x = (f_+)_x - (f_-)_x$ with both parts ν-integrable, so $f_x \in L^1(\nu)$; thus $f_x \in L^1(\nu)$ for almost every x. Also, for $x \in E$, $\varphi(x) = \varphi_1(x) - \varphi_2(x)$, and since $\varphi_1, \varphi_2 \in L^1(\mu)$, $\varphi \in L^1(\mu)$. Finally, applying Tonelli to f_+ and f_-,

$$\int_X \varphi\,d\mu = \int_X \varphi_1\,d\mu - \int_X \varphi_2\,d\mu = \int_{X\times Y} f_+\,d(\mu\times\nu) - \int_{X\times Y} f_-\,d(\mu\times\nu) = \int_{X\times Y} f\,d(\mu\times\nu).$$

This proves half of the theorem; the other half is identical. ■

Exercise 4.2.3

In Fubini's theorem let Y be the set of natural numbers with the measure of Fact B.4 (counting measure). What conclusion do you get? What if X is also taken to be $\mathbb{N}$ in the same way?

Solution. With $Y = \mathbb{N}$ and counting measure c, a function on $X \times \mathbb{N}$ is a sequence $f_n(x) = f(x,n)$ of measurable functions, and $\int_{\mathbb{N}} g\,dc = \sum_n g(n)$. Fubini says: if $\sum_n \int_X |f_n|\,d\mu < \infty$ (equivalently $\int_X \sum_n |f_n|\,d\mu < \infty$, by Tonelli), then $\sum_n f_n(x)$ converges absolutely for a.e. x, defines an integrable function, and

$$\int_X \sum_{n=1}^{\infty} f_n\,d\mu = \sum_{n=1}^{\infty} \int_X f_n\,d\mu.$$

If also $X = \mathbb{N}$ with counting measure, $f(m,n) = a_{mn}$ is a double sequence and we get: if $\sum_m \sum_n |a_{mn}| < \infty$, then

$$\sum_{m=1}^{\infty}\sum_{n=1}^{\infty} a_{mn} = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} a_{mn},$$

both sides converging absolutely; Tonelli gives the same equality for $a_{mn} \geq 0$ without any condition. □

Exercise 4.2.4

On $\mathbb{N} \times \mathbb{N}$ consider

$$f(m,n) = \begin{cases} 2 - 2^{-m}, & m = n, \\ -2 + 2^{-m}, & m = n+1, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the two iterated sums and see how this relates to the hypotheses of Tonelli's and Fubini's theorems. Do the same for

$$f(x,y) = \frac{x^2 - y^2}{(x^2+y^2)^2}, \qquad x, y \in [1,\infty).$$

Solution. (i) Fix n and sum over m: only $m = n$ and $m = n+1$ contribute, giving $(2 - 2^{-n}) + (-2 + 2^{-n-1}) = -2^{-n-1}$. Hence $\sum_n \sum_m f(m,n) = -\sum_{n\geq 1} 2^{-n-1} = -\frac{1}{2}$. Fix m and sum over n: for $m = 1$ only $n = 1$ contributes, giving $2 - \frac{1}{2} = \frac{3}{2}$; for $m \geq 2$, $n = m$ and

$n = m - 1$ give $(2 - 2^{-m}) + (-2 + 2^{-m}) = 0$. Hence $\sum_m \sum_n f(m,n) = \frac{3}{2}$. The iterated sums differ. This does not contradict the theorems: f takes both signs, so Tonelli does not apply, and $\sum\sum|f(m,n)| \geq \sum_n (2 - 2^{-n}) = \infty$, so the hypothesis of Fubini fails.

(ii) Since $\dfrac{\partial}{\partial y}\dfrac{y}{x^2+y^2} = \dfrac{x^2-y^2}{(x^2+y^2)^2}$,

$$\int_1^\infty f(x,y)\,dy = \Big[\frac{y}{x^2+y^2}\Big]_{y=1}^{\infty} = -\frac{1}{1+x^2}, \qquad \int_1^\infty\int_1^\infty f\,dy\,dx = -\int_1^\infty \frac{dx}{1+x^2} = -\frac{\pi}{4}.$$

Since $f(y,x) = -f(x,y)$, the other order gives $+\frac{\pi}{4}$. Again f changes sign and $\iint|f| = \infty$ (otherwise Fubini would force equality), so neither theorem applies. □

4.2.1 Borel functions

A function on $\mathbb{R}^n$ is a **Borel function** if it is measurable with respect to $\mathcal{B}_n$; plain "measurable" means measurable with respect to Lebesgue measure $(\mathcal{M}_n, m_n)$. Borel functions are of course measurable. The following extends Fact B.26(d).

Proposition 4.2.4

If $f : X \to \mathbb{R}$ is measurable with respect to $(X, \mathcal{S}, \mu)$ and $g : \mathbb{R} \to \mathbb{R}$ is a Borel function, then $g \circ f : X \to \mathbb{R}$ is measurable.

Proof. For each $\alpha \in \mathbb{R}$, $g^{-1}(\alpha,\infty) \in \mathcal{B}_1$, so it suffices to show $f^{-1}(E) \in \mathcal{S}$ for every $E \in \mathcal{B}_1$. Let

$$\mathcal{A} = \{E \in \mathcal{B}_1 : \ f^{-1}(E) \in \mathcal{S}\}.$$

$\mathcal{A}$ is a σ-algebra of subsets of $\mathbb{R}$ (Exercise 4.2.5), and every open set lies in $\mathcal{A}$ (an open set is a countable union of open intervals, and $f^{-1}(a,b) = \{f > a\} \cap \{f < b\} \in \mathcal{S}$). Hence $\mathcal{A} = \mathcal{B}_1$. ■

Exercise 4.2.5

In the proof of Proposition 4.2.4, show that $\mathcal{A}$ is a σ-algebra.

Solution. $f^{-1}(\emptyset) = \emptyset \in \mathcal{S}$. If $E \in \mathcal{A}$ then $f^{-1}(E^c) = X \setminus f^{-1}(E) \in \mathcal{S}$. If $E_n \in \mathcal{A}$ then $f^{-1}(\bigcup_n E_n) = \bigcup_n f^{-1}(E_n) \in \mathcal{S}$. Since $\mathcal{B}_1$ is itself a σ-algebra, these sets also lie in $\mathcal{B}_1$. □

To apply Tonelli or Fubini to a function of two variables we must know it is $\mathcal{M} \times \mathcal{M}$-measurable, which is tedious to verify directly. Every Borel function on $\mathbb{R}^2$ is $\mathcal{M} \times \mathcal{M}$-measurable (as $\mathcal{B}_2 \subset \mathcal{M} \times \mathcal{M}$), so the next proposition is useful; it also yields Tonelli–Fubini for arbitrary measurable functions of two variables.

Proposition 4.2.5

For every measurable function f on $\mathbb{R}^n$ there is a Borel function g with $f = g$ almost everywhere.

Proof. It suffices to treat $f \geq 0$. By Fact B.6 write $f(x) = \sum_{k=1}^\infty c_k \chi_{E_k}(x)$ with $c_k \geq 0$ and $E_k \in \mathcal{M}_n$. For each k there are Borel sets $A_k \subset E_k \subset B_k$ with $m_n(B_k \setminus A_k) = 0$. Put

$$g(x) = \sum_{k=1}^{\infty} c_k \chi_{A_k}(x), \qquad x \in \mathbb{R}^n.$$

By Fact B.26, g is a Borel function. Moreover

$$\{x : f(x) \neq g(x)\} \subset \bigcup_{k=1}^{\infty} (E_k \setminus A_k) \subset \bigcup_{k=1}^{\infty} (B_k \setminus A_k),$$

a null set, so $f = g$ almost everywhere. ■

Theorem 4.2.6 (Tonelli–Fubini on $\mathbb{R}^2$)

Let f be a measurable function on $\mathbb{R}^2$.

(a) For almost every x and almost every y, f_x and f^y are measurable.

(b) If $0 \le f \le \infty$, then

$$\int_{\mathbb{R}}\int_{\mathbb{R}} f(x,y)\,dx\,dy = \int_{\mathbb{R}\times\mathbb{R}} f\,dm_2 = \int_{\mathbb{R}}\int_{\mathbb{R}} f(x,y)\,dy\,dx. \tag{9}$$

(c) If $f:\mathbb{R}^2\to\mathbb{C}$ and $\int_{\mathbb{R}}\int_{\mathbb{R}}|f(x,y)|\,dx\,dy<\infty$, then $f\in L^1(m_2)$; moreover $f_x, f^y\in L^1(m_1)$ for almost every x,y, and (9) holds.

Proof. By Proposition 4.2.5, $f=g+h$ where g is a Borel function and h is measurable with $h=0$ almost everywhere. As discussed above, g is $\mathcal{M}\times\mathcal{M}$-measurable, so g satisfies everything in the theorem by Proposition 4.2.1 and Theorems 4.2.2, 4.2.3 (note $\int g\,dm_2 = \int g\,d(m\times m)$ since m_2 extends $m\times m$). It therefore suffices to prove the statements for h. Let

$$E=\{(x,y)\in\mathbb{R}^2 : h(x,y)\neq 0\}.$$

Then $E\in\mathcal{M}_2$ and $m_2(E)=0$, so there is $F\in\mathcal{M}\times\mathcal{M}$ with $E\subset F$ and $(m\times m)(F)=0$. By (8), $\int_{\mathbb{R}} m(F_x)\,dx=(m\times m)(F)=0$, so

$$A=\{x\in\mathbb{R} : m(F_x)>0\}$$

has $m(A)=0$. Fix $x\in\mathbb{R}\setminus A$. Then $m(F_x)=0$ and $E_x\subset F_x$, so $m(E_x)=0$ (by completeness of m). If $y\notin E_x$ then $h_x(y)=h(x,y)=0$; so h_x vanishes outside a null set, hence is measurable and $h_x=0$ a.e. Thus for almost every x, h_x is measurable and $h_x(y)=0$ for almost every y. The same holds for h^y, proving (a); (b) and (c) for h follow at once since all integrals of h, h_x, h^y (for a.e. x,y) vanish. ∎

Exercise 4.2.6

Complete the proof of Theorem 4.2.6.

Solution. Write $f=g+h$ as in the proof. For (b), let $f\ge 0$; we may take $g\ge 0$ (replace g by g_+, still Borel, and h by $f-g_+$, still zero a.e.). For $x\notin A$, $f_x=g_x$ a.e., so $\int f_x\,dy=\int g_x\,dy$ for a.e. x, hence $\int\!\int f\,dy\,dx=\int\!\int g\,dy\,dx$. Also $\int f\,dm_2=\int g\,dm_2=\int g\,d(m\times m)$ because $f=g$ m_2-a.e. and g is $\mathcal{M}\times\mathcal{M}$-measurable with $m_2=m\times m$ there. Tonelli for g gives $\int\!\int g\,dy\,dx=\int g\,d(m\times m)=\int\!\int g\,dx\,dy$, and the $dx\,dy$ side for f is handled by the symmetric argument with h^y. This is (9). For (c), $|f|$ is measurable and nonnegative, so by (b) $\int|f|\,dm_2=\int\!\int|f|\,dx\,dy<\infty$, i.e. $f\in L^1(m_2)$; then $g\in L^1(m\times m)$ and Fubini applies to g. Since $f_x=g_x$ a.e. for a.e. x (and $f^y=g^y$ a.e. for a.e. y), the integrability of the sections and equality (9) transfer from g to f. □

Example 1 (the Dirichlet integral)

Apply Fubini to $f(x,t)=e^{-xt}\sin x$ on $[0,A]\times[0,\infty)$. Since

$$\frac{1}{x}=\int_0^\infty e^{-xt}\,dt, \qquad x>0,$$

we have

$$\int_0^1\!\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \le \int_0^1 x\int_0^\infty e^{-xt}\,dt\,dx \le \int_0^1 dx = 1,$$
$$\int_1^A\!\int_0^\infty |e^{-xt}\sin x|\,dt\,dx \le \int_1^A\!\int_0^\infty e^{-xt}\,dt\,dx = \int_1^A \frac{1}{x}\,dx < \infty,$$

so Fubini applies (using $|\sin x| \le x$ on $[0,1]$). Therefore

$$\int_0^A \frac{\sin x}{x}\,dx = \int_0^A \int_0^\infty e^{-xt}\sin x\,dt\,dx = \int_0^\infty \int_0^A e^{-xt}\sin x\,dx\,dt$$
$$= \int_0^\infty \Big[\frac{e^{-At}}{1+t^2}(-t\sin A - \cos A) + \frac{1}{1+t^2}\Big]dt.$$

Since

$$\int_0^\infty \Big|\frac{e^{-tA}}{1+t^2}(-t\sin A - \cos A)\Big|dt \le \int_0^\infty 2e^{-tA}\,dt = \frac{2}{A}$$

(using $|t\sin A + \cos A| \le 1 + t$ and $\frac{1+t}{1+t^2} \le 2$), we conclude

$$\lim_{A\to\infty}\int_0^A \frac{\sin x}{x}\,dx = \int_0^\infty \frac{dt}{1+t^2} = \frac{\pi}{2}.$$

4.3 Convolution of Integrable Functions

Among the Banach spaces $L^p(X,\mu)$ $(1 \le p \le \infty)$, $L^1(X,\mu)$ plays a special role: if X carries an operation and μ has a property like Fact B.1 (translation invariance), one can define a product on $L^1(X,\mu)$. Here we treat only Lebesgue integration on the real line.

4.3.1 Definition

For measurable $f, g : \mathbb{R} \to \mathbb{C}$ and $x \in \mathbb{R}$ consider

$$\int_{\mathbb{R}} f(y)g(x-y)\,dy.$$

By Proposition 4.2.5 we may assume f, g are Borel functions without changing this integral. The maps $(x,y) \mapsto y$ and $(x,y) \mapsto x - y$ are continuous, so by Proposition 4.2.4 the functions $(x,y) \mapsto f(y)$ and $(x,y) \mapsto g(x-y)$ are Borel, and by Fact B.26(a) their product

$$(x,y) \mapsto f(y)g(x-y) \tag{10}$$

is also a Borel function. (When f, g are merely measurable, (10) is still a measurable function of two variables.) Now

$$\int_{\mathbb{R}}\Big[\int_{\mathbb{R}}|f(y)g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}}|f(y)|\Big[\int_{\mathbb{R}}|g(x-y)|\,dx\Big]dy = \int_{\mathbb{R}}|f(y)|\,dy\int_{\mathbb{R}}|g(x)|\,dx = \|f\|_1\|g\|_1,$$

using translation invariance (Fact B.1). By Fubini, the function

$$x \mapsto \int_{\mathbb{R}} f(y)g(x-y)\,dy \tag{11}$$

is defined almost everywhere, and its L^1-norm satisfies

$$\int_{\mathbb{R}}\Big|\int_{\mathbb{R}} f(y)g(x-y)\,dy\Big|dx \le \int_{\mathbb{R}}\Big[\int_{\mathbb{R}}|f(y)g(x-y)|\,dx\Big]dy = \|f\|_1\|g\|_1.$$

Definition (convolution)

For $f, g \in L^1(\mathbb{R})$ the L^1-function (11) is denoted $f * g$ and called the **convolution** of f and g. We have just proved

$$\|f * g\|_1 \le \|f\|_1 \|g\|_1, \qquad f, g \in L^1(\mathbb{R}). \tag{12}$$

Proposition 4.3.1

For $f, g, h \in L^1(\mathbb{R})$ and constants $\alpha, \beta \in \mathbb{C}$:

(a) $[(f * g) * h](x) = [f * (g * h)](x)$ for almost every x;

(b) $(f * g)(x) = (g * f)(x)$ for every x (for which either side is defined);

(c) $[(\alpha f + \beta g) * h](x) = [\alpha(f * h) + \beta(g * h)](x)$ for almost every x.

Proof. (a) Fix $x \in \mathbb{R}$. Then

$$[(f * g) * h](x) = \int_{\mathbb{R}} (f * g)(y) h(x - y)\, dy = \int_{\mathbb{R}} \Big[\int_{\mathbb{R}} f(z) g(y - z)\, dz \Big] h(x - y)\, dy,$$

and

$$\begin{aligned} [f * (g * h)](x) &= \int_{\mathbb{R}} f(z)(g * h)(x - z)\, dz = \int_{\mathbb{R}} f(z) \Big[\int_{\mathbb{R}} g(y) h(x - z - y)\, dy \Big] dz \\ &= \int_{\mathbb{R}} f(z) \Big[\int_{\mathbb{R}} g(y - z) h(x - y)\, dy \Big] dz \end{aligned}$$

(substituting $y \mapsto y - z$). Now

$$\int_{\mathbb{R}} \int_{\mathbb{R}} |f(z) g(y - z) h(x - y)|\, dy\, dz = [(|f| * |g|) * |h|](x),$$

and since $|f| * |g|$ and $|h|$ are L^1-functions, this value is finite for almost every x. Fix such an x and apply Fubini to $(y, z) \mapsto f(z) g(y - z) h(x - y)$; this proves (a).

(b) Directly,

$$(f * g)(x) = \int_{\mathbb{R}} f(y) g(x - y)\, dy = \int_{\mathbb{R}} f(-y) g(x + y)\, dy = \int_{\mathbb{R}} f(x - y) g(y)\, dy = (g * f)(x),$$

where the second equality uses $m(E) = m(-E)$ (reflection invariance) and the third uses translation invariance.

(c) For almost every x the functions $y \mapsto f(y) h(x - y)$ and $y \mapsto g(y) h(x - y)$ are L^1-functions, so Fact B.10 (linearity of the integral) gives (c). ∎

Definition (Banach algebra)

A Banach space X with a multiplication $(x, y) \mapsto xy$ that is associative, distributive and satisfies

$$\|xy\| \le \|x\| \|y\|, \qquad x, y \in X,$$

is called a **Banach algebra**. By (12) and Proposition 4.3.1(a),(c), $L^1(\mathbb{R})$ is a Banach algebra under convolution.

Proposition 4.3.2

If g is differentiable and $f, g, g' \in L^1(\mathbb{R})$, then $f * g$ is differentiable and

$$(f * g)' = f * g'.$$

Exercise 4.3.1

Prove Proposition 4.3.2.

Solution. *Step 1: g is bounded and $g(b) - g(a) = \int_a^b g'$.* A function differentiable everywhere with integrable derivative satisfies the fundamental theorem of calculus (see Ch. 5), so $g(x) = g(0) + \int_0^x g'$. Since $g' \in L^1$, the limits $g(\pm\infty)$ exist, and since $g \in L^1$ they are 0; hence g is bounded and $(f*g)(x)$ is defined for *every* x.

Step 2: an integral formula. For $x \in \mathbb{R}$, $h \neq 0$,

$$(f * g)(x + h) - (f * g)(x) = \int_{\mathbb{R}} f(y) \int_0^h g'(x - y + t)\, dt\, dy = \int_0^h (f * g')(x + t)\, dt,$$

by Fubini, which applies because $\int_{\mathbb{R}} \int_0^h |f(y)g'(x - y + t)|\, dt\, dy \leq |h|\, \|f\|_1 \|g'\|_1 < \infty$.

Step 3. Hence $\dfrac{(f * g)(x + h) - (f * g)(x)}{h}$ is the average of $f * g'$ over the interval between x and $x + h$. Whenever $f * g'$ is continuous at x (for instance at every x if g' is bounded, since then $f * g'$ is continuous by Exercise 4.3.2), this tends to $(f * g')(x)$, which gives $(f * g)' = f * g'$. In general Step 2 shows $f * g$ is absolutely continuous with $(f * g)' = f * g'$ almost everywhere (Lebesgue differentiation, Ch. 5).

Alternative under bounded g'. If $|g'| \leq M$, the mean value theorem gives $\left|f(y)\frac{g(x+h-y)-g(x-y)}{h}\right| \leq M|f(y)|$, an integrable bound, and the difference quotient converges pointwise to $f(y)g'(x - y)$; the dominated convergence theorem gives $(f * g)'(x) = (f * g')(x)$ for every x. □

4.3.2 Convolution of L^p and L^q functions

Let $p, q > 1$ be conjugate exponents. For a function g and $s \in \mathbb{R}$ write $g_s(t) = g(t - s)$. If $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$, Hölder's inequality gives

$$\int_{\mathbb{R}} |f(y)g(x - y)|\, dy = \int_{\mathbb{R}} |f(y)g_{-x}(-y)|\, dy \leq \|f\|_p \|g_{-x}\|_q = \|f\|_p \|g\|_q.$$

Hence $(f * g)(x)$ is defined for every $x \in \mathbb{R}$ and $\|f * g\|_\infty \leq \|f\|_p \|g\|_q$.

Exercise 4.3.2

If $f \in L^1(\mathbb{R})$ and $g \in L^\infty(\mathbb{R})$, show that $f * g$ is uniformly continuous.

Solution. Using $f * g = g * f$, for $x, y \in \mathbb{R}$,

$$\begin{aligned} |(f * g)(x) - (f * g)(y)| &= \left| \int_{\mathbb{R}} g(z)(f(x - z) - f(y - z))dz \right| \\ &\leq \|g\|_\infty \int_{\mathbb{R}} |f(x - z) - f(y - z)|\, dz = \|g\|_\infty \|f_{x-y} - f\|_1, \end{aligned}$$

after the substitution $w = y - z$. By the continuity of translation in L^1 (Fact B.23), $\|f_s - f\|_1 \to 0$ as $s \to 0$. Hence for $\epsilon > 0$ there is $\delta > 0$ with $|(f * g)(x) - (f * g)(y)| < \epsilon$ whenever $|x - y| < \delta$. □

Proposition 4.3.3

Let $p, q \in (1, \infty)$ be conjugate exponents. Then $f * g \in C_0(\mathbb{R})$ for every $f \in L^p(\mathbb{R})$ and $g \in L^q(\mathbb{R})$.

Proof. By Hölder's inequality again, for $x, y \in \mathbb{R}$

$$|(f * g)(x) - (f * g)(y)| \le \int_{\mathbb{R}} |f(z)(g_{-x}(-z) - g_{-y}(-z))|\, dz \le \|f\|_p \|g_{-x} - g_{-y}\|_q,$$

so by Fact B.23 (continuity of translation in L^q, $q < \infty$), $f * g$ is continuous. Next, by Fact B.22 there are sequences $\langle f_n \rangle, \langle g_n \rangle$ in $C_c(\mathbb{R})$ with

$$\lim_{n\to\infty} \|f_n - f\|_p = 0, \qquad \lim_{n\to\infty} \|g_n - g\|_q = 0.$$

If the supports of f_n and g_n lie in $[-A, A]$ and $[-B, B]$, then the support of $f_n * g_n$ lies in $[-A - B, A + B]$, so $\langle f_n * g_n \rangle$ is a sequence in $C_c(\mathbb{R})$. Then

$$\begin{aligned} \|f_n * g_n - f * g\|_\infty &\le \|f_n * g_n - f_n * g\|_\infty + \|f_n * g - f * g\|_\infty \\ &\le \|f_n\|_p \|g_n - g\|_q + \|f_n - f\|_p \|g\|_q \to 0, \end{aligned}$$

so $\langle f_n * g_n \rangle$ converges uniformly to $f * g$. A uniform limit of functions in $C_c(\mathbb{R})$ belongs to $C_0(\mathbb{R})$. ■

4.3.3 Averaging kernels

It is usually hard to guess what $f * g$ looks like, but sometimes one can. Consider

$$k_r = \frac{1}{2r} \chi_{[-r,r]}, \qquad r > 0. \tag{13}$$

By definition

$$(k_r * f)(x) = \int_{-\infty}^{\infty} k_r(x - y) f(y)\, dy = \frac{1}{2r} \int_{x-r}^{x+r} f(y)\, dy,$$

the average of f over the neighbourhood $[x - r, x + r]$ of x. So $k_r * f$ should be better behaved than f and approach f as $r \to 0$.

$k_r * f$ is uniformly continuous for $f \in L^1(\mathbb{R})$. By Fact B.15 (absolute continuity of the integral), for any $\epsilon > 0$ there is $\delta \in (0, 2r)$ with

$$E \in \mathcal{M},\ m(E) \le \delta \implies \int_E |f(t)|\, dt < 2r\epsilon.$$

If $|x - y| < \frac{\delta}{2} < r$, let $E = [x - r, x + r] \triangle [y - r, y + r]$, where $A \triangle B = (A \cup B) \setminus (A \cap B)$; then $m(E) = 2|x - y| \le \delta$. Therefore

$$|(k_r * f)(x) - (k_r * f)(y)| = \frac{1}{2r} \Big| \int_{x-r}^{x+r} f(t)\, dt - \int_{y-r}^{y+r} f(t)\, dt \Big| \le \frac{1}{2r} \int_E |f(t)|\, dt < \epsilon,$$

so $f * k_r$ is uniformly continuous on $\mathbb{R}$.

Uniform approximation. If f is uniformly continuous, for each $\epsilon > 0$ there is $\delta > 0$ with $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$. If $r < \delta$, then

$$|f(x) - (f * k_r)(x)| \le \frac{1}{2r} \int_{x-r}^{x+r} |f(x) - f(t)|\, dt < \epsilon,$$

so $f * k_r \to f$ uniformly as $r \to 0$.

Exercise 4.3.3

For $f = \chi_{[-1,1]}$ sketch $x \mapsto (f * k_r)(x)$ for $r = 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{10}$. Do the same for $f(x) = \max\{1 - |x|, 0\}$.

Solution. *(i)* $f = \chi_{[-1,1]}$. $(f * k_r)(x) = \frac{1}{2r}\, m([x - r, x + r] \cap [-1, 1])$, an even function. For $r \le 1$:

$$(f * k_r)(x) = \begin{cases} 1, & |x| \le 1 - r, \\ \dfrac{1 + r - |x|}{2r}, & 1 - r \le |x| \le 1 + r, \\ 0, & |x| \ge 1 + r. \end{cases}$$

This is a trapezoid: flat top of height 1 on $[-(1 - r), 1 - r]$ and linear ramps of width $2r$ centred at ± 1, passing through the value $\frac{1}{2}$ at $x = \pm 1$. For $r = 1$ it is the triangle $(2 - |x|)/2$ on $[-2, 2]$; as $r = \frac{1}{2}, \frac{1}{3}, \frac{1}{5}, \frac{1}{10}$ the ramps steepen and the graph approaches the step $\chi_{[-1,1]}$. The convergence is pointwise (to $\frac{1}{2}$ at ± 1), not uniform: f is not continuous.

(ii) $f(x) = \max\{1 - |x|, 0\}$ *(tent).* For $r \le \frac{1}{2}$ and $x \ge 0$ (the function is even), a direct integration gives

$$(f * k_r)(x) = \begin{cases} 1 - \dfrac{x^2 + r^2}{2r}, & 0 \le x \le r, \\ 1 - x, & r \le x \le 1 - r, \\ \dfrac{(1 + r - x)^2}{4r}, & 1 - r \le x \le 1 + r, \\ 0, & x \ge 1 + r. \end{cases}$$

The tent's corners at 0 and ± 1 are replaced by parabolic arcs of width $2r$; the result is C^1, its maximum is $1 - \frac{r}{2}$, and $\|f * k_r - f\|_\infty = \frac{r}{2} \to 0$ (uniform convergence, since f is uniformly continuous). For $r = 1$: $(f * k_1)(x) = \frac{1}{2} - \frac{x^2}{4}$ for $|x| \le 1$ and $\frac{(2 - |x|)^2}{4}$ for $1 \le |x| \le 2$, a flatter bell with peak $\frac{1}{2}$. □

Exercise 4.3.4

Let $f_0 = \chi_{[-1,1]}$ and $f_{n+1} = f_n * f_0$ $(n = 0, 1, 2, \dots)$. Sketch how the graphs of this sequence change.

Solution. $f_1(x) = m([x - 1, x + 1] \cap [-1, 1]) = \max\{2 - |x|, 0\}$, a triangle of height 2 on $[-2, 2]$. Inductively, since $(f_n * f_0)(x) = \int_{x-1}^{x+1} f_n$:

- f_n is even, nonnegative, supported in $[-(n + 1), n + 1]$, increasing on $(-(n + 1), 0]$;
- f_n is a piecewise polynomial of degree n with breakpoints at the integers of the same parity as $n + 1$, and $f_n \in C^{n-1}$ (each convolution with f_0 gains one derivative: $f'_{n+1}(x) = f_n(x + 1) - f_n(x - 1)$);
- $\int f_n = \left(\int f_0\right)^{n+1} = 2^{n+1}$ by Fubini.

Thus the graphs go step $\to$ triangle $\to$ piecewise quadratic bell $\to \cdots$, becoming smoother and wider. After normalizing, $f_n / 2^{n+1}$ is the density of a sum of $n + 1$ independent uniform variables on $[-1, 1]$, which has variance $(n + 1)/3$; by the central limit theorem the graphs look increasingly like the Gaussian curve $\frac{2^{n+1}}{\sqrt{2\pi(n+1)/3}} e^{-3x^2/(2(n+1))}$. □